

Multi-turn Conversational AI from Text to Multimodal Interaction: Data, Models, Evaluation, and Open Challenges

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Abstract

Conversational AI is moving beyond isolated text prompts toward sustained, multimodal interaction. In real conversations, users clarify goals, revise requests, interrupt responses, switch topics, and introduce new evidence while expecting systems to preserve context across turns. This makes multi-turn dialogue a distinct challenge requiring systems to maintain and update memory, ground responses across modalities, tools, and external knowledge, and adapt across languages and cultures. This study reviews **multi-turn conversational AI** across text-only dialogue, AudioLLMs and speech-native systems, multimodal and omni-modal systems, and tool-augmented agents. We organize the literature around *datasets* and *benchmarks*, *modeling paradigms*, *training strategies*, *evaluation setups*, and *cross-cutting challenges*. Our analysis shows that support for multiple modalities has advanced faster than the ability to sustain coherent interaction across a session. Despite stronger capabilities to perceive, speak, and act across modalities, current systems still struggle with persistent memory, cross-turn grounding, full-duplex interaction, robust evaluation, and cultural alignment. We conclude with a research agenda for systems that can remember, revise, ground, speak, listen, act, and adapt across turns, modalities, and cultures.¹

1 Introduction

Conversational AI is becoming a primary interface through which users interact with large language models (LLMs). These interactions often extend beyond single prompts. Users clarify goals, revise requests, ask follow-up questions, switch topics, introduce new evidence, and return to earlier points in the conversation (Kwan et al., 2024; Bai et al., 2024; Reddy et al., 2019). We use *session-level multi-turn interaction* to refer to a complete

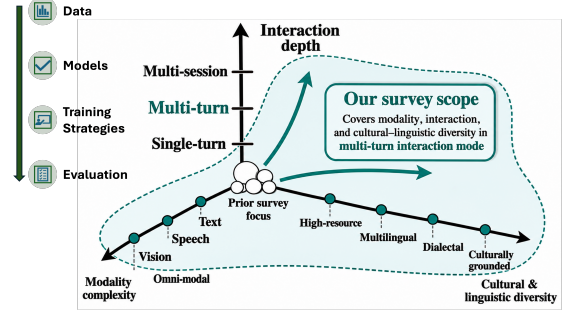


Figure 1: Three-axis view of conversational AI covering modality complexity, interaction depth, and cultural-linguistic diversity.

conversation with more than one exchange, where later turns may depend on earlier requests, responses, evidence, actions, or state. A capable model must therefore do more than answer the current query. It must preserve context, resolve cross-turn references, update earlier assumptions, and maintain coherence across the session. Multi-turn dialogue is thus a distinct modeling and evaluation problem, not merely a longer version of single-turn question answering (Deshpande et al., 2025).

This problem has become more important as conversational AI expands beyond text. Early dialogue systems focused on text-based state tracking, response generation, persona consistency, and task completion (Wu et al., 2020; Zhang et al., 2020b; Roller et al., 2021). AudioLLMs extend this setting to spoken interaction by processing speech with text (Zhang et al., 2023; Chu et al., 2024; Tang et al., 2024). Full-duplex systems support more flexible turn-taking, including real-time interruption and overlapping speech (Défossez et al., 2024). More recent omni-modal models jointly handle text, speech, vision, and often video within unified architectures (Hurst et al., 2024; Xu et al., 2025b; Fu et al., 2026; Chen et al., 2025c; Tong et al., 2025). Figure 2 illustrates this progression. Each stage broadens what systems can perceive and produce, while making memory, grounding, timing, and alignment across turns more difficult.

¹[multiturn-conversational-resources](#)

Recent evidence shows that even frontier models still struggle with sustained interaction. Models can underuse relevant information even when it remains inside the context window (Liu et al., 2024a). They also degrade when a task is distributed across several turns rather than stated as a complete single-turn instruction (Laban et al., 2026). MultiChallenge further shows that frontier models remain below human-level reliability on realistic multi-turn instruction following (Deshpande et al., 2025). These failures become more complex in spoken, multimodal, and tool-augmented settings. Speech recognition or acoustic-token errors can propagate across turns, visual grounding can decay as the dialogue moves away from the original image or video, and tool-using agents must preserve external state in addition to dialogue history. Evaluation setups that score turn-level responses can therefore miss failures that only emerge at the session level (Kwan et al., 2024; Graf et al., 2026).

Related Surveys. Prior surveys provide useful coverage of dialogue systems, multi-turn LLM capabilities, agents, and multimodal LLMs, however, they study these areas largely in isolation. Dialogue-system surveys review the transition from modular or retrieval-based systems to LLM-based dialogue, however, remain largely text-centered (Wang et al., 2023; Yi et al., 2025). Surveys on multi-turn interaction discuss instruction following, memory, and consistency, however, do not cover spoken or multimodal interaction in depth (Zhang et al., 2025b; Li et al., 2025f). Work on LLM agents focuses mainly on tool use and agent evaluation (Guan et al., 2026), while surveys of multimodal LLMs often treat dialogue as a downstream application rather than the central unit of analysis (Zhang et al., 2024b). Cultural and dialectal variation is also largely absent from these discussions. In contrast, this survey treats **sustained interaction** as the *primary unit of analysis* across modalities, systems, resources, and evaluation methods. Table 1 summarizes how our scope differs from the closest related surveys. Appendix Table 11 provides a more detailed comparison of their conceptual framing, modality coverage, and key distinctions.

Survey Scope and Selection. We organize our coverage along three dimensions: *interaction depth* (multi-turn sessions where later responses depend on earlier context), *modality complexity*

Survey	MT	Speech	Vision	Tri	Cultural	Eval Δ
(YI ET AL., 2025)	✓	×	×	×	×	✓
(WANG ET AL., 2023)	✓	×	×	×	×	×
(ZHANG ET AL., 2025B)	✓	×	×	×	×	✓
(LI ET AL., 2025F)	✓	×	×	×	•	✓
(GUAN ET AL., 2026)	✓	×	×	×	×	✓
(ZHANG ET AL., 2024B)	×	•	✓	×	×	×
This survey	✓	✓	✓	✓	✓	✓

Table 1: Comparison with closely related surveys. **MT** = multi-turn focus; **Speech** = spoken dialogue/AudioLLMs; **Vision** = image/video; **Tri** = joint text–speech–vision; **Cultural** = cross-lingual, dialectal, or culturally grounded coverage; **Eval Δ** = multi-turn evaluation gaps. • denotes partial coverage.

(text through speech to omni-modal systems), and *cultural and linguistic diversity* (multilingual, dialectal, and culturally grounded settings). Figure 1 illustrates this joint space. We include work that advances dialogue across turns or develops the **data, models, training methods**, and **evaluation** needed for sustained interaction, spanning topics such as *text dialogue, spoken dialogue, AudioLLMs, vision-language and video dialogue, omni-modal systems, conversational retrieval, tool use, and culturally grounded resources*. Details keywords are provided in Appendix Sec. A.4. To ensure a systematic, and reproducible review of the literature, we followed *PRISMA-ScR* framework (Page et al., 2021). To maximize coverage, we searched Google Scholar and Semantic Scholar and obtained papers from IEEE Xplore, ACM Digital Library, Elsevier, DBLP, the ACL Anthology and arXiv. Papers include publications from *ACL, NeurIPS, ICLR, ICML, Interspeech, SIGDIAL, CVPR, ICCV, AAAI and related venues. Our initial search identified ~4K papers. After applying the PRISMA screening, we selected 200 papers for detailed review.

Contributions. This survey makes following contributions.

- We provide a unified survey of multi-turn conversational AI spanning text, speech, vision, video, agentic, and culturally grounding.
- We organize datasets, benchmarks, models, training strategies, and evaluation methods around session-level interaction rather than isolated responses.
- We identify open challenges in memory, coherence, paralinguistic understanding, multi-modal grounding, real-time interaction, and cultural alignment.

Findings. This survey highlights several gaps in the current multi-turn conversational AI.

- **Capability gap.** Longer context windows, stronger base models, and broader modality support do not guarantee session-level competence. Systems still struggle with memory, grounding, assumption revision, tool state, spoken timing, and cross-cultural adaptation.
- **Resource gap.** Existing datasets remain dominated by English with (over 80%), text-only, and text-image interaction. Speech, video, omni-modal, and culturally grounded multi-turn resources remain limited.
- **Evaluation gap.** Benchmarks increasingly use LLM-as-judge and mixed approaches, but session-level metrics, human validation, and reproducible scoring remain underdeveloped.
- **Integration gap.** Most resources test one dimension at a time. Few combine long-horizon memory, multilinguality, speech, visual grounding, tool use, safety, and cultural alignment in the same evaluation setting.

2 Multi-turn Dialogue

We define a *conversation* as an interaction episode between a user and a system, consisting of ordered turns. A *multi-turn conversation* is a session with $T > 1$ exchanges, where later turns may depend on earlier requests, responses, evidence, actions, or state. This dependency makes the full session the natural unit of analysis rather than a single response. We define a session, $\mathcal{D}$, as

$$\mathcal{D} = \{(u_t, y_t, c_t, m_t, z_t)\}_{t=1}^T, \\ y_t = f_\theta(u_t, c_t, m_t, z_t).$$

Here u_t is the user input at turn t , y_t is the system response, c_t is the accumulated dialogue context from earlier turns, m_t is the multimodal context, and z_t is optional external state. The multimodal context may include speech, images, video, or other non-text signals, and may be empty in text-only settings. The external state may include memory, retrieved evidence, tool outputs, task state, or user profile information. The function f_θ denotes the conversational model. This formulation makes the multi-turn setting explicit because each response can depend on the current input, the previous dialogue, modality-specific context, and external state.

Modality and Context. The user input u_t and system response y_t may be expressed as *text*,

speech, *images*, *video*, or *their combination*. The multimodal context m_t captures modality specific information such as acoustic cues, visual content, or video events. The dialogue context c_t captures what has already happened in the session, while the external state z_t captures information that may persist or change. Since these signals carry different information, the model must decide what to retain, update, or ignore at each turn.

Dialogue Types. Dialogue systems serve different interaction functions. We consider **task oriented dialogue** for goal completion via state tracking (Wu et al., 2020; Eric et al., 2020), **open-domain dialogue** for coherent conversation across topics (Zhang et al., 2020b; Roller et al., 2021), **knowledge grounded dialogue** for responses grounded in documents, databases, or retrieval (Varshney et al., 2022), **social and empathetic dialogue** for socially appropriate responses (Zhang et al., 2024d), **persona grounded dialogue** for consistency with a user or character profile (Gosling et al., 2023; Oh et al., 2023), **agentic dialogue** for planning and tool use (Yao et al., 2022; Lu et al., 2024), and **multimodal grounded dialogue** for reasoning over text, speech, images, video, or their combination (Xue et al., 2025). These types often overlap, so they serve as analytical lenses not strict dataset boundaries.

Evaluation Focus. A multi-turn system is mainly evaluated at the **turn** and **session** levels. Turn level evaluation asks whether the current response is correct and useful. Session level evaluation asks whether the system remains consistent, grounded, and useful as earlier turns shape later ones. Some settings add further requirements. Persistent memory extends the problem beyond one session, tool using systems add external actions and state changes, and spoken systems may require full duplex evaluation for interruptions, overlapping speech, and response timing. Our focus is interaction within a multi-turn session, while cross session memory and agentic interaction are covered as related extensions when they affect session level behavior.

3 Datasets and Benchmarks

Datasets and benchmarks for multi-turn dialogue differ not only in modality, but also in what they assume a system must preserve across turns. Text resources emphasize context tracking, instruction

retention, persona consistency, retrieval, and social intelligence. Spoken resources add acoustic cues, turn-taking, prosody, speaker variation, ASR errors, and full-duplex timing. Multimodal and video resources add visual, audio-visual, and temporal grounding. Cultural and linguistic resources expose whether these abilities transfer beyond English-centered settings.

We organize datasets by modality in Table 2 and benchmarks by evaluation focus in Table 3. Overall, they cover text-only, spoken, multimodal, video, and culturally grounded interaction. Some findings are as follows.

- **English and text remain dominant.** Outside the cultural and cross-lingual block, 43 of 52 dataset entries are English-only. The benchmark table shows the same pattern, with 32 of 53 benchmarks focusing on text-only dataset.
- **Modality coverage is expanding, however, varies across modalities.** Table 2 lists 18 text-image resources, however, only 8 spoken and 6 video-oriented datasets. Many multimodal resources still use static images rather than streaming, temporal, or spoken interaction.
- **Task coverage is broadening.** Though QA and instruction following dominate, newer resources add memory, personalization, tool use, retrieval, social interaction, and emotional intelligence. Long-horizon and multi-session dialogue remain rare.
- **Cultural coverage is growing, however, often single-turn.** Resources such as CVQA, Dallah, MMA-ASIA, OASIS, and Shawarma Chats improve cultural coverage. However, 4 of the 5 entries in this block are single-turn and do not test sustained interaction.
- **Evaluation needs stronger calibration.** LLM-as-judge and mixed scoring now dominate many benchmarks. These setups scale well, but require stronger human evaluation and clearer reproducibility. Safety and robustness evaluation also remains text-centered.

4 Modeling Paradigm

Modeling paradigms have shifted from modular dialogue pipelines to general-purpose LLMs, AudioLLMs, omni-modal systems, and tool-augmented agents, as shown in Figure 2. Appendix Table 5 summarizes the details. This evolution has broadened the interaction interface, however, it has not solved the central multi-turn chal-

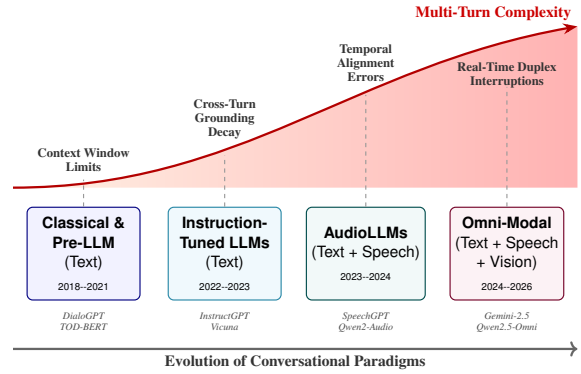


Figure 2: Evolution of conversational architectures. Multi-turn challenges escalate as systems evolve from simple text to complex omni-modal agents.

lenge. Most systems still treat dialogue history as a flat context, while only a smaller body of work explicitly models memory, state updates, selective grounding, and efficient long-session reasoning.

Classical systems separated language understanding, dialogue state tracking, policy learning, and response generation, often using rule-based, statistical, or neural sequence-to-sequence components. **Transformer-era** dialogue models began to consolidate these functions through pretraining, multi-task learning, knowledge injection, and explicit memory mechanisms.

Instruction-tuned LLMs then shifted dialogue modeling away from task-specific managers toward general-purpose models trained or adapted for instruction following, open-ended interaction, and structured task behavior through prompting or function-like abstractions. The next stage added speech, vision, and real-time interaction.

AudioLLMs extend conversational AI from text-only interaction to spoken dialogue by modeling speech alongside language. Early systems represent speech as discrete tokens within an LLM framework, enabling speech input and output to be handled as part of the same sequence modeling problem. Later models broaden this direction toward chat-ready audio understanding, generic hearing, and interleaved speech-text modeling.

A second line targets real-time spoken interaction, supporting streaming generation, controllable voice, emotion, timbre, and full-duplex dialogue with low latency. Textless spoken dialogue models further show that dialogue can be modeled directly from raw audio, preserving turn-taking and paralinguistic cues often lost in transcripts.

Overall, AudioLLMs move dialogue systems toward speech-native interaction. Yet most still handle multi-turn context through the underlying lan-

Reference	Mod.	Task	Turns	Size	Lang.	Split	Curation
Text-only							
PERSONACHAT (Zhang et al., 2018)	T	PS	6–20	10,981/164K ^u	en	Tr	H
CoQA (Reddy et al., 2019)	T	QA	6–20	8,399	en	Tr+Ev	H
MULTIWOZ2.1 (Eric et al., 2020)	T	TOD/DST	6–20	10,438	en	Tr+Ev	H
PIPPA (Gosling et al., 2023)	T	PS	> 20	25,940/>1M ^u	en	Tr	H+S
ULTRACHAT (Ding et al., 2023)	T	Inst	≤5	1.5M	en	Tr	S
MT-BENCH (Zheng et al., 2023)	T	Inst	≤5	80 ^u	en	Ev	H
WILDCHAT (Zhao et al., 2024b)	T	OD	≤5	1M	ML(68)	Tr+Ev	H
MT-BENCH-101 (Bai et al., 2024)	T	Inst	≤5	1,388	en	Ev	H+S
MT-EVAL (Kwan et al., 2024)	T	Inst	6–20	168	en	Ev	H+S
PRODIGY (Occhipinti et al., 2024)	T	PS	4	20,850	en	Tr+Ev	H+S
LONGMEMEVAL (Wu et al., 2025b)	T	Mem	MS	500 ^u	en	Ev	H+S
MULTICHALLENGE (Deshpande et al., 2025)	T	Inst	≤10 / 5 avg	273	en	Ev	H+S
LMSYS-CHAT-1M (Zheng et al., 2024)	T	OD	≤5	1M	ML(154)	Tr+Ev	H+S
τ-BENCH (Yao et al., 2025)	T	TOD	—	165tasks	en	Ev	S+H
PERSONAMEM (Jiang et al., 2025)	T	Pers/Mem	MS	180+ histories / ~6K ^u	en	Ev	S+H
CONSISTENTCHAT (Chen et al., 2025a)	T	Inst	6–20	15K / 224K ^u	en	Tr	S
DOCTALK (Lee et al., 2025a)	T	KG/QA	>20	730K	en	Tr	H+S
TOOLWOZ (Lattimer et al., 2025)	T	TOD+Tool	—	7,849 scenarios	en	Tr+Ev	H+S
SOTOPIA (Zhou et al., 2024a)	T	Social/PS	6–20	450 tasks	en	Ev	H+S
DIALSIM (Kim et al., 2024b)	T	Mem	> 20	18.99K turns / 1.31K sess.	en	Ev	S+H
Spoken (audio / speech)							
SPOKENWOZ (Si et al., 2023)	T+S	TOD/DST	> 20	5.7K / 249 ^h	en	Tr+Ev	H
DEEPDIALOGUE (Koudounas et al., 2025)	T+S	OD/EI	3–10	40,150	en	Tr	S + H
AUDIO MULTICHALLENGE (Gosai et al., 2025)	S	Inst	6–20	452	en	Ev	H+S
MENASPEECHBANK (Ali et al., 2026)	T+S	TOD/Pers	≤5	417K	ar*	Tr+Ev	H+S
C3 (Ma et al., 2025)	T+S	QA	6–20	1,079	en+zh	Ev	H+S
ASK-QA (Chen et al., 2025d)	T+S	QA	≤5	7,830	en	Tr+Ev	S
MULTI-BENCH (Deng et al., 2025)	S	OD+EI	6–20	1.5K	en+zh	Ev	H+S
MSIB (Tong et al., 2025)	T+S	OD	6–20	244	en	Ev	S+H
Multimodal							
VisDIAL (Das et al., 2017)	T+V	QA	6–20	~123K / 1.23M ^u	en	Tr+Ev	H
MMDIALOG (Feng et al., 2023a)	T+V	OD	≤5	1.08M	en	Tr	H
IMAD (Moskvoretskii et al., 2024)	T+V	OD	6–20	4864	en	Tr	H
INFOVisDIAL (Wen et al., 2023)	T+V	KG	6–20	50K	en	Tr	S
DIALOGCC (Lee et al., 2024)	T+V	OD	6–20	83K	en	Tr	H+S
LoCoMo (Maharana et al., 2024)	T+V	Mem	>20	10 / 1,986 ^u	en	Ev	H+S
TMDIALOG (Lei et al., 2025)	T+V	Inst	6–20	67.9K / 329	en	Tr+Ev	H+S
MMDU-45K (Liu et al., 2024d)	T+V	Inst	6–20	45K / 410K ^u	en	Tr	S+H
CONVBENCH (Liu et al., 2024b)	T+V	Inst	≤5	577	en	Ev	H+S
CB-300K (Tian et al., 2025b)	T+V	QA	≤5	340k/717K ^u	en	Tr+Ev	H+S
MMDIAG (Liu et al., 2025a)	T+V	QA	≤5	639K ^u	en	Tr+Ev	H+S
MULTIVERSE (Lee et al., 2025b)	T+V	Inst	≤5	647	en	Ev	H+S
MMRC (Xue et al., 2025)	T+V	Mem+QA	6–20	5,120 / 28K ^u	en	Ev	H
ALIGNMMBENCH (Wu et al., 2025d)	T+V	Inst	—	4,978 ^u	zh	Ev	H+S
MEM-GALLERY (Bei et al., 2026)	T+V	Mem	MS	240 sess. / 1,711 ^u	en	Ev	H+S
MMMB (Tong et al., 2025)	T+V	Mem	6–20	300	en	Ev	S+H
DIALOGBEN (Huang et al., 2025b)	T+V	T2I	≤5	9,957	en+zh	Ev	S
MMMT-IF (Epstein et al., 2024)	T+V	Inst	1–20	71	en	Ev	H+S
Video (text + video / streaming)							
MT-VIDEO-BENCH (Pan et al., 2025)	T+Vid	QA	6–20	1,000 / 5,887 ^u	en	Ev	H+S
OMNIMMI (Wang et al., 2025c)	T+S+Vid	QA	≤5	1,121 ^u / 2,290 ^u	en	Ev	H+S
SCVBENCH (You et al., 2025)	T+Vid	QA	6–20	925 ^u / 7,280 ^u	en	Ev	H+S
COGSTREAM (Zhao et al., 2026b)	T+Vid	QA	—	1,088 ^u / 59,032 ^u	en	Tr+Ev	H+S
IVCR-200K (Han et al., 2024)	T+Vid	Ret	6–20	12,516 ^u / 201,631 ^u	en+zh	Tr+Ev	S+H
SVBENCH (Yang et al., 2025)	T+Vid	QA	≤5	49,979 ^u / 1,353 ^u	zh	Tr+Ev	H+S
Cultural (* single-turn)							
CVQA* (Mogrovejo et al., 2024)	T+V	QA	1	10,374 ^u	ML(31)	Ev	H
DALLAH* (Alwajih et al., 2024)	T+V	QA	1	20×6 ^u	ar(6D)	Ev	H
MMA-ASIA* (Weihua et al., 2025)	T+S+V	QA	1	27,000 ^u	ML(10)	Ev	H+S
OASIS* (Alam et al., 2025)	T+S+V	QA	1	14.8M ^u	en+ar*	Tr+Ev	H+S
SHAWARMA CHATS (Zeinalipour et al., 2025)	T	OD+KG	6–20	30K	ar*(3D)	Tr+Ev	H+S

Table 2: Multi-turn dialogue **datasets** organised by modality. **Mod.:** T=text, S=speech/audio, V=image/vision, Vid=video. **Task:** TOD=task-oriented dialogue, OD=open-domain, QA=question answering, PS=persona / roleplay, Pers.=personalization / user profiling, KG=knowledge-grounded, Inst=instruction-following benchmark, T2I=text-to-image generation / editing, EI=emotional intelligence, Ret=retrieval, Mem=long-term memory. **Turns:** approximate range per session (≤5 / 6-20 / >20 / MS=multi-session). **Size:** # dialogues unless marked: ^u=utterances; ^q=QA pairs / instances; ^h=hours; ^v=videos. **Split:** Tr=training data, Ev=evaluation data, Tr+Ev=both. **Curation:** H=human; S=synthetic / LLM-generated; **Lang.:** ML(*)=Multilingual; Number in parenthesis indicate number of languages. D=Dialects. Datasets marked * are *single-turn*; they are included solely as cultural / cross-lingual baselines.

guage model, with limited explicit modeling of long-horizon spoken memory, interruptions, and cross-turn acoustic grounding.

Omni-modal models extend AudioLLMs by jointly handling text, speech, vision, and sometimes video. Recent systems explore several design choices, including streaming duplex inter-

action, low-latency speech and vision alignment, shared multimodal tokenization, modality-specific encoders, and separate modules for reasoning and speech generation. These designs move conversational AI from speech-text interaction toward unified perception and generation across modalities.

Audio-visual dialogue forms a more specialized

Benchmark	Mod.	Turns	Eval
General multi-turn instruction-following & consistency			
MT-BENCH (Zheng et al., 2023)	T	2	J
MT-BENCH-101 (Bai et al., 2024)	T	3	J
MT-EVAL (Kwan et al., 2024)	T	7	M
MULTICHALLENGE (Deshpande et al., 2025)	T	5 avg / ≤ 10	M
IHEVAL (Zhang et al., 2025e)	T	2	R
TURNWISE (Graf et al., 2026)	T	2–8	J
TOD-PROCBENCH (Ghazarian et al., 2025)	T	–	R
EVOLIF (Jia et al., 2025)	T	–	M
PARROT-BENCH (Sun et al., 2024)	T	8	J
τ -BENCH (Yao et al., 2025)	T	–	M
STRUCTFLOWBENCH (Li et al., 2025b)	T	4.14 avg	J
PERSONAMEM (Jiang et al., 2025)	T	≤ 60 sessions	Ac
CORAL (Cheng et al., 2025)	T	8.26 avg	M
TOOLSANDBOX (Lu et al., 2025)	T	13.9 avg	rM
TURNBENCH-MS (Zhang et al., 2025d)	T	>20	M
MINT (Wang et al., 2024)	T	≤ 5	R
SOTOPIA (Zhou et al., 2024a)	T	≤ 20	M
AGENTBOARD (Ma et al., 2024)	T	3–25	M
MEMORYAGENTBENCH (Hu et al., 2025)	T	–	M
Multilingual & cross-lingual			
M2LINGUAL (Maheshwary et al., 2025)	T	2	J
CMT-EVAL (Tian et al., 2025a)	T	multi	J
ALIGNMMBENCH (Wu et al., 2025d)	T+V	– avg	M
MT-BENCH-Hi (Kamath et al., 2025)	T	2	J
CUDIALOG (Cao et al., 2024)	T	8 sent. (5+3)	R
INDOTOD (Kautsar et al., 2023)	T	2.63/4.06 avg	R
Multimodal (image + text)			
CONVBENCH (Liu et al., 2024b)	T+V	3	J
MMDU (Liu et al., 2024d)	T+V	15 avg / 27 max	J
MMCR (Yan et al., 2025a)	T+V	4/8	J
MMRC (Xue et al., 2025)	T+V	\$15.2 avg	M
MULTIVERSE (Lee et al., 2025b)	T+V	3.91 avg	J
MEM-GALLERY (Bei et al., 2026)	T+V	MS	M
MMMT-IF (Epstein et al., 2024)	T+V	1–20	M
MMMB (Tong et al., 2025)	T+V	≤ 15	J
Spoken & video			
AUDIO MULTICHALLENGE (Gosai et al., 2025)	S	3–8	M
MT-VIDEO-BENCH (Pan et al., 2025)	T+Vid	5–8	J
OMNIMMI (Wang et al., 2025c)	T+S+Vid	stream / 1–3	M
SCVBENCH (You et al., 2025)	T+Vid	6.16 avg	R
COGSTREAM (Zhao et al., 2026b)	T+Vid	stream; 5.02	J
AVHBENCH (Sung-Bin et al., 2025)	T+S+V	–	M
MTALK-BENCH (Du et al., 2025b)	S	2–3	M
FD-BENCH (Peng et al., 2025)	S	≤ 5	M
MULTI-BENCH (Deng et al., 2025)	S	8 avg	M
SVBENCH (Yang et al., 2025)	T+Vid	stream; 4.29	J
MSIB (Tong et al., 2025)	S	2–10	M
Robustness, fairness & safety			
FB-BENCH (Li et al., 2025d)	T	2	M
FAIRMT-BENCH (Fan et al., 2025)	T	multi	M
SYCON-BENCH (Hong et al., 2025)	T	multi	R
CURSE OF MULTI-MODALITIES (Leng et al., 2026)	T+S+V	–	M
LOST IN MULTI-TURN (Laban et al., 2026)	T	multi	M
X-TEAMING (Rahman et al., 2025)	T	avg 5.10	M
CRESCENDO (Rusinovich et al., 2025)	T	1–7	R
DIAHALU (Chen et al., 2024b)	T	6.91 avg	H+J
SAFEDIALBENCH (Cao et al., 2026)	T	3–10	M

Table 3: Multi-turn evaluation benchmarks grouped by modality and evaluation focus. **Mod.** denotes modality, with T for text, S for speech, V for vision, and Vid for video. **Turns** reports the average number of turns per instance or session when available, and otherwise gives a qualitative range. **Eval** denotes the evaluation protocol, with R for rule-based scoring, H for human evaluation, J for LLM-as-judge, M for mixed evaluation, and Ac for accuracy. Benchmarks that evaluate only single-turn ability are excluded.

direction in which visual cues support speaker tracking, turn-taking, and grounding rather than serving only as general perceptual input. AV-Dialog uses lip-centered visual features to identify the target speaker, predict turn transitions,

and generate responses under noise and competing speech (Chen et al., 2026). MAViD instead combines multimodal understanding with synchronized audio-video response generation through a Conductor–Creator architecture, though it is not evaluated for sustained multi-turn interaction (Pang et al., 2025). Despite this progress, most omni-modal systems still treat dialogue history as a flat sequence. Multi-turn-aware methods address grounded memory, context management, and long-session inference cost, while long-horizon grounding, reference tracking, and session-level coherence remain open challenges.

Agentic dialogue systems extend multi-turn modeling by framing conversation as a loop of planning, tool use, observation, and revision. ReAct interleaves reasoning and actions (Yao et al., 2022), Reflexion uses feedback and episodic memory to improve later attempts (Shinn et al., 2023), and MetaGPT enables role-based collaboration through structured workflows (Hong et al., 2024). Recent systems apply these ideas to tool use, function calling, web navigation, recommendation, video reasoning, and live multimodal interaction (Wu et al., 2024; Lu et al., 2024). These settings introduce three key challenges. First, agents must track external states and dependencies that may not appear in the dialogue history; ToolSandBox evaluates such stateful and implicit dependencies. Second, tool calls can produce undesirable or irreversible effects, motivating evaluation of both task completion and harmful side effects. Third, failed actions require replanning and backtracking. Reflexion and SCoRe support self-correction, but do not provide transactional rollback for already executed external actions. Overall, modeling has progressed from modular state tracking to unified, multimodal, and agentic systems, while long-horizon grounding and cross-turn memory remain key open challenges. Appendix A.2 provides additional model-level details.

5 Training Strategies

Multi-turn dialogue training must account for dependencies across turns, delayed rewards, changing user intent, and context-sensitive response quality. We group existing training approaches into five families: (i) supervised fine-tuning, (ii) reinforcement learning and preference optimization, (iii) multi-task learning, (iv) synthetic data generation, and (v) conversational retrieval-augmented

training. In Appendix Table 5, we summarize modeling paradigms, while Table 6 presents the main training strategies. Table 7 further compares these different families in terms of their mechanisms, suitable settings, strengths, limitations, and modality coverage.

Supervised fine-tuning. Supervised fine-tuning requires conversations that reflect real multi-turn phenomena such as follow-up questions, anaphora, ellipsis, topic shifts, and safety escalation. UltraChat and WildChat provide large-scale synthetic and real multi-turn chat data, while Parrot explicitly targets referential phenomena in follow-up turns. Domain-specific resources such as Aquila-Med, Qilin-Med, and Zhongjing adapt SFT to multi-turn clinical dialogue, and XGuard-Train extends SFT to multi-turn adversarial safety training.

Reinforcement learning and preference optimization. Multi-turn reinforcement learning shifts the learning signal from isolated responses to full interactions. InstructGPT establishes the standard RLHF pipeline, while ArCHer and MT-RLHF adapt optimization to longer conversations and conversation-level preferences. Preference methods such as DMPO, Multi-turn DPO/KTO, Parrot, and SDPO optimize preferences at response, segment, or trajectory level. Other work trains action selection, tool use, self-correction, proactive interaction, and long-horizon sparse-reward behavior across turns. Clinical and tutoring settings further combine SFT, RLHF, DPO, expert feedback, and multi-agent RL for specialized multi-turn interaction.

Multi-task learning. Multi-task learning improves transfer by sharing representations across dialogue objectives. PPTOD jointly trains response generation, dialogue state tracking, and policy prediction, while TOD-BERT learns dialogue-aware representations across task-oriented dialogue corpora. DAMSEL shows that auxiliary objectives are useful for conversational QA when labeled speech data is limited.

Synthetic data generation. Synthetic generation is widely used to scale multi-turn data and control dialogue structure. UltraChat, MMDU-45K, MMDiag, and TMDiag generate instruction-following, multimodal, note-taking, and context-modeling dialogues. Other pipelines target Arabic dialogue, emotional speech, tool use, coherent intent trajectories, multi-topic information seeking, reviewer-guided regeneration, task-oriented bootstrapping, and multi-turn API/tool-use simulation.

Conversational RAG. Conversational RAG trains models to retrieve, filter, and use evidence across turns. PK-ICR and commonsense retrieval approaches ground responses in persona, external knowledge, and knowledge graphs. ChatQA, ChatQA-2, HAConvDR, UniConv, IterCQR, and related work improve conversational retrieval, query reformulation, history selection, and joint retrieval-generation training. KEDiT and CORAL further explore efficient evidence integration and citation-aware conversational RAG training. To translate this comparison into practice, in Table 8, we map common deployment goals to the training strategy. In Appendix A.2, we provide additional details on training strategies.

6 Evaluation of Multi-turn Dialogue

Evaluating multi-turn dialogue requires moving beyond single-response metrics. A response may be fluent and factually correct on its own; however, it can still fail at the session level if it ignores earlier instructions, loses user preferences, misuses retrieved evidence, mishandles tool state, or breaks grounding across modalities. Evaluation therefore needs to measure both local response quality and cross-turn consistency, memory, state tracking, grounding, interaction success, and robustness across the full dialogue.

Existing evaluation methods fall into five broad groups. (i) Surface-form and task-oriented metrics, such as overlap scores, semantic similarity, dialogue state tracking, and task success, remain useful, though they provide limited insight into session-level coherence. (ii) Session-level metrics assess instruction retention, memory recall, constraint satisfaction, feedback integration, and dialogue-level hallucination. (iii) Agentic and retrieval-grounded evaluation measures tool use, stateful execution, evidence retrieval, and source attribution. (iv) Speech-native and full-duplex evaluation adds speech quality, paralinguistic cues, interruption handling, latency, and response timing. (v) LLM-as-a-judge and human evaluation support open-ended assessment; however, their reliability depends on rubric quality and may be affected by judge bias, rater inconsistency, and small ranking differences. In Table 10, we compare these five families by what they measure, the systems they best suit, and their main limitations, showing that no single family fully captures session-level competence. Appendix A.3

provides a detailed discussion of these approaches, and Table 9 summarizes representative metrics and frameworks.

Evaluation gaps. Current setups make multi-turn failures more visible, however, several gaps remain. (i) Strong single-turn performance does not reliably transfer to session-level interaction. (ii) Agentic dialogue lacks unified evaluation for hidden state, tool-side effects, source attribution, and repeated-trial reliability. (iii) Spoken and full-duplex systems still lack integrated session-level evaluation that jointly measures semantic correctness, audio quality, timing, interruptions, and paralinguistic behaviour. (iv) Multi-turn evaluation remains focused in English and high-resource settings, leaving multilingual, dialectal, and culturally grounded evaluation split across separate resources. These gaps show that current evaluation still measures many components of dialogue competence separately, rather than testing reliable session-level behaviour across languages, modalities, tools, and time.

7 Challenges, and Future Directions

The studied literature shows progress in datasets, modeling, training, and evaluation, however, sustained multi-turn interaction remains unresolved. Stronger single-turn models alone do not achieve multi-turn competence. Systems must preserve context, revise assumptions, ground responses, coordinate tools, and maintain quality across long, multimodal, and culturally diverse sessions. Mechanisms for memory, cross-turn grounding, speech-native interaction, robust revision, and session-level evaluation remain limited.

Persistent memory and state management.

Current long-context models can process more dialogue history, however, they still struggle to retrieve, update, and apply the right information across turns. This problem grows in multi-session settings, where preferences, facts, and task states change over time. Future work should move beyond flat context windows toward explicit memory systems that separate dialogue state, and task-specific working memory. These systems should support updates, forgetting, conflict resolution, and transparent retrieval across sessions.

Cross-turn grounding. Grounding weakens when responses depend on earlier turns, external evidence, visual content, acoustic cues, or tool

outputs. Full dialogue history can add noise in retrieval-augmented dialogue, while multimodal references often decay across turns. Future work should develop selective grounding methods that identify the relevant dialogue history, evidence, and modality streams for each turn. This requires structured dialogue state rather than treating the full history as a flat input sequence.

Speech-native and full-duplex interaction.

Spoken dialogue adds timing, prosody, interruptions, repairs, noise, and paralinguistic cues that transcript-only pipelines often miss. Full-duplex systems make this harder because users and systems may speak at the same time. Future work should treat speech as an interaction medium, not only as an input modality. This requires training and evaluation for interruption handling, response timing, spoken clarification, voice-grounded tool use, and paralinguistic understanding.

Robustness and revision. Multi-turn systems often commit to early assumptions and fail to revise them after later corrections. They are also vulnerable to adversarial escalation, where individually benign turns accumulate into unsafe or incorrect outcomes. Future work should improve session-level robustness by enabling models to detect uncertainty, recover from mistakes, and maintain safety throughout the interaction.

Evaluation mismatch. Current evaluation methods capture only parts of multi-turn behavior. Surface-form metrics, task-oriented metrics, agentic benchmarks, speech-native evaluation settings, and LLM-as-judge methods remain difficult to compare. No unified framework yet measures session-level competence across memory, grounding, tool use, speech, safety, and user satisfaction. Future evaluation should move from turn-level scoring to session-level assessment, with explicit measures of state consistency, evidence use, revision, timing, and repeated-trial reliability.

Cultural and linguistic coverage. Most resources and evaluations still focus on English and high-resource settings. Dialects, code-switching, cultural pragmatics, and region-specific knowledge shape how users express intent and judge appropriate responses. Future benchmarks should move beyond isolated language-specific datasets toward comparable multilingual and culturally grounded evaluation suites. This is especially im-

portant for multimodal settings, where language, culture, accent, and visual context interact.

Beyond turn-by-turn exchange. Most systems still follow a simple user-query and assistant-response loop. This framing does not fully capture proactive suggestions, mid-utterance clarification, streaming perception, emotion-aware interaction, or mixed-initiative collaboration. Future systems need interaction models that treat dialogue as continuous, stateful, and adaptive rather than as a sequence of independent turns.

8 Conclusion

Multi-turn conversational AI is shifting from text-only dialogue toward sustained interaction across speech, vision, video, tools, and culturally diverse contexts. This study shows that modality support has advanced faster than session-level competence. Current systems can increasingly perceive, speak, and act, but still struggle to maintain memory, revise assumptions, preserve grounding, coordinate external state, handle spoken timing, and adapt across languages and cultures. We argue that future work should evaluate conversational systems not only by the quality of isolated responses, but by their ability to sustain coherent, grounded, safe, and culturally appropriate interaction across turns.

Limitations

This study focuses on recent work on multi-turn conversational AI across data, models, training, and evaluation. While we aim to cover major directions, the literature is growing quickly, and some new or concurrent resources may not be included. We cover diverse systems into broad categories, which may hide finer differences in architecture, training data, deployment setting, or evaluation setups.

Societal/Broader Impact

This study aims to clarify the capabilities, gaps, and risks of multi-turn conversational AI. By organizing prior work across datasets, models, training, and evaluation, it can help researchers design systems that better support long-horizon interaction, multilingual access, speech-based interfaces, multimodal assistance, and culturally grounded communication.

Ethical Considerations

This study does not introduce new datasets or run experiments with human participants.

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A Appendix

A.1 Additional Resources

A.1.1 Text-only Multi-turn Resources

Foundational benchmarks. CoQA (Reddy et al., 2019) contains conversational question answering grounded in source passages and emphasizes dependencies across previous turns. MultiWOZ 2.1 (Eric et al., 2020) provides multi-domain task-oriented dialogues with corrected state annotations and is widely used for dialogue state tracking, inform rate, and task success. PersonaChat (Zhang et al., 2018) pairs open-domain

conversations with explicit persona profiles, making it a standard resource for persona-consistency research.

Capability evaluation. MT-Bench-101 (Bai et al., 2024) introduces a hierarchical taxonomy for fine-grained multi-turn evaluation. MT-Eval (Kwan et al., 2024) evaluates recollection, expansion, refinement, and follow-up, showing that models can degrade when a task is distributed across turns. MultiChallenge (Deshpande et al., 2025) targets realistic human-LLM conversations and evaluates instruction retention, inference memory, reliable versioned editing, and self-coherence. MINT (Wang et al., 2024) restructures reasoning, code, and decision-making tasks into a user-tool-LLM interaction format.

Long-term memory. LoCoMo (Maharana et al., 2024) evaluates long-term conversational memory through extended multi-session dialogues grounded in personas and temporal event graphs. LongMemEval (Wu et al., 2025b) tests information extraction, multi-session reasoning, knowledge updates, temporal reasoning, and abstention. DialSim (Kim et al., 2024b) evaluates long-term multi-party dialogue using TV-series transcripts and response-time constraints. PersonaMem (Jiang et al., 2025) evaluates dynamic user profiling and preference tracking across multi-session histories. Mem-Gallery (Bei et al., 2026) extends long-memory evaluation to multimodal settings.

Training corpora. UltraChat (Ding et al., 2023) provides synthetic multi-turn instruction-following conversations. WildChat (Zhao et al., 2024b) contains opt-in real ChatGPT conversations with broad topic and language diversity. LMSYS-Chat-1M (Zheng et al., 2024) collects real user conversations with multiple LLMs through public chat interfaces. DocTalk (Lee et al., 2025a) synthesizes multi-turn information-seeking dialogues from Wikipedia and is mainly designed as a pre-training resource.

Persona, role-play, and social interaction. PIPPA (Gosling et al., 2023) contains long role-play conversations and supports evaluation of in-character consistency. PRODIGy (Occhipinti et al., 2024) grounds personas in fictional character profiles and studies the effect of backstory and speaking style. SOTOPIA (Zhou et al., 2024a)

evaluates social intelligence through role-play scenarios where agents pursue social goals while maintaining appropriate behavior.

Tool use, retrieval, and proactive assistance.

τ -Bench (Yao et al., 2025) evaluates agents in stateful user-tool interaction settings with task and policy constraints. ToolSandbox (Lu et al., 2025) evaluates stateful tool use and interaction consistency across turns. mtRAG (Katsis et al., 2025) evaluates multi-turn retrieval-augmented generation, where the agent must decide when and what to retrieve as the conversation evolves. ProMiSe (Butala et al., 2024) evaluates proactive information seeking by asking agents to generate useful follow-up suggestions during a dialogue.

A.1.2 Spoken Dialogue Resources

Foundational resources. TurnGPT (Ekstedt and Skantze, 2020) applies a Transformer language model to turn-taking prediction in spoken dialogue. It frames turn completion as a language-modeling problem over transcribed conversational text, providing an early trainable baseline for predicting when a speaker’s turn is likely to end. SpokenWOZ (Si et al., 2023) provides 249 hours of human-human task-oriented speech across eight domains and 203K turns. It preserves audio, transcripts, and dialogue-state annotations, allowing evaluation of how recognition errors affect downstream state tracking and task success.

Multi-turn speech-to-speech evaluation. Audio MultiChallenge (Gosai et al., 2025) extends multi-turn evaluation to spoken dialogue systems using 452 conversations and 1,712 instance-level rubrics. It evaluates the four axes of MultiChallenge together with a Voice Editing axis for mid-utterance speech repairs, and compares turn-level and conversation-level scores. URO-Bench (Yan et al., 2025b) evaluates end-to-end speech-to-speech dialogue models across multilingual, multi-round, and paralinguistic settings, with basic and pro tracks covering 20 S2S task types. MTalk-Bench (Du et al., 2025b) constructs multi-turn speech-to-speech test dialogues through an LLM-human pipeline, then adds human recording, voice conversion, and ambient sound mixing. It evaluates semantic content, vocal cues, and background sound using both arena-style pairwise comparison and rubric-based absolute scoring. MULTI-Bench (Deng et al., 2025) focuses on emotional intelligence in spoken dialogue and tests whether

models can track and respond to emotional-state changes across turns. FD-Bench (Peng et al., 2025) evaluates simultaneous bidirectional speech through simulated five-round conversations, measuring interruption handling and response timing with metrics such as SIR, SRIR, EIR, NIR, IRD, FSED, ERT, and EIT, as referred in Table 9.

Related audio-agent benchmarks.

ADU-Bench (Gao et al., 2025a) provides open-ended audio dialogue tasks covering multiple scenarios, skills, languages, and ambiguity types such as intonation and homophones. Because it is not explicitly multi-turn in the current categorization, it is best treated as a related audio-dialogue benchmark rather than a core multi-turn resource. Full-Duplex-Bench-v3 (Lin et al., 2026) extends full-duplex benchmarking to tool-use scenarios, measuring whether voice agents invoke tools correctly while handling disfluencies such as false starts and repairs. Since it is marked as work in progress, it can be mentioned as an emerging direction rather than a central benchmark.

Training corpora and specialized spoken resources.

DeepDialogue (Koudounas et al., 2025) provides synthetic multi-turn dialogues paired with emotionally consistent speech, making it useful for training spoken dialogue models with emotion-control signals. ASK-QA (Chen et al., 2025d) contains spoken conversational QA sessions with multiple speakers and deliberately ambiguous user requests, supporting research on clarification and spoken question answering. The MENASpeechBank (Ali et al., 2026) provides persona-conditioned conversations in Arabic and regional dialects across the MENA region, combining voice-bank design with multi-turn spoken Arabic dialogue.

A.1.3 Multimodal Dialogue Resources

Foundational and open-domain resources.

VisDial (Das et al., 2017) pairs COCO images with ten rounds of question answering and established the standard format for multi-turn visual dialogue. MMDialog (Feng et al., 2023a) provides large-scale real-world multimodal conversations with images across thousands of topics and introduces MM-Relevance for modality-aware response evaluation. DialogCC (Lee et al., 2024) and IMAD (Moskvoretskii et al., 2024) construct image-augmented dialogues by matching images to existing text conversations using

vision-language similarity, making them scalable resources for multimodal dialogue training. InfoVisDial (Wen et al., 2023) adds Wikipedia-grounded knowledge to visual dialogue, requiring models to combine visual evidence with external factual information across turns.

Instruction-tuning corpora. MMDU-45K (Liu et al., 2024d) contains multi-turn, multi-image dialogues generated with GPT-4o and human refinement, with sessions that can include many images and long image-text contexts. MMCR (Yan et al., 2025a) provides both an instruction-tuning corpus and a diagnostic benchmark covering multiple domains and subtopics, supporting evaluation under single-image and multi-image conversational settings.

Capability evaluation. ConvBench (Liu et al., 2024b) evaluates multi-turn LVLM dialogues across perception, reasoning, and creation tasks, showing that early visual perception errors can propagate into later turns. MultiVerse (Lee et al., 2025b) evaluates multi-turn multimodal dialogue across many task types and explicitly tests in-context learning during conversation. MMRC (Xue et al., 2025) evaluates information extraction, multi-turn reasoning, information update, image management, memory recall, and refusal behavior, identifying recurring failures such as memory degradation, update failure, and error propagation. MMT-IF (Epstein et al., 2024) focuses on multi-turn instruction following with image inputs, making it a multimodal counterpart to text-only instruction-retention benchmarks. Chatterbox/CB-300K (Tian et al., 2025b) targets multi-round referring and grounding, with emphasis on referential ambiguity, spatial relations, and grounding consistency across turns.

Long-term memory and context management. Mem-Gallery (Bei et al., 2026) evaluates multi-session multimodal memory across visual and textual histories, including extraction, reasoning, and knowledge management. DiagNote/MM-Diag (Liu et al., 2025a) provides multi-turn dialogues with explicit note-taking annotations and studies whether learned notes can help VLMs maintain grounded context over long interactions.

Emerging tri-modal, generative, cultural, and video resources. InteractiveOmni (Tong et al., 2025) extends multi-turn dialogue to tri-modal

audio-visual-language interaction, using cross-modal context and historical memory. DialogGen/DialogBen (Huang et al., 2025b) evaluates dialogue-conditioned image generation and editing, where the model must preserve both conversational context and output-modality coherence. AlignMMBench (Wu et al., 2025d) evaluates cultural alignment and factual consistency in single- and multi-turn Chinese and bilingual multimodal dialogues. SVBench (Yang et al., 2025) introduces temporal multi-turn dialogue over streaming video, where later questions depend on both earlier answers and specific video segments.

A.1.4 Cultural and Linguistic Resources

Arabic and dialectal multi-turn resources. Shawarma Chats (Zeinalipour et al., 2025) provides 30K six-turn Arabic conversations grounded in Wikipedia content, covering MSA, Egyptian Arabic, and Maghrebi Arabic. It is one of the few large-scale multi-turn resources with explicit coverage of Maghrebi Arabic, making it useful for dialect-aware dialogue modeling and evaluation. Naous et al. (2020) introduced an empathy-driven Arabic conversational chatbot and an accompanying message-response corpus spanning Levantine, Egyptian, and Gulf Arabic, providing an early resource for Arabic dialectal conversation modeling. Alexandria (El Mekki et al., 2026) is a multi-turn conversational machine translation dataset for English-Dialectal Arabic, covering 13 Arab countries and 11 domains with city-level dialect metadata and speaker-addressee gender information. It supports evaluation of dialect naturalness, style and formality, semantic accuracy, and gender-sensitive translation.

Cross-lingual and culturally grounded multi-turn benchmarks. C3 (Ma et al., 2025) is a bilingual English-Chinese spoken dialogue benchmark targeting phonological ambiguity, coreference, omission, and multi-turn dependency. Kamath et al. (2025) introduce a Hindi evaluation suite consisting of MT-Bench-Hi for multi-turn instruction following, ChatRAG-Hi for conversational retrieval-augmented generation, and BFCL-Hi for function calling. cuDialog (Cao et al., 2024) conditions response prediction on Hofstede cultural-value dimensions, making cultural behavior an explicit variable in multi-turn dialogue evaluation. IndoToD (Kautsar et al., 2023) provides an Indonesian multi-domain task-oriented

Resource	Year	Languages	Region / Cultures	MT	MM	Dialect	Venue
<i>Arabic / MENA</i>							
OASIS / EVERYDAYMMQA (Alam et al., 2025)	2025	EN, MSA, Egyptian, Levantine	18 Arab countries	✗*	✓	✓	arXiv
M ² CQA (Mousi et al., 2026)	2025	MSA + dialects	17 MENA countries	✗	✓	✓	arXiv
DALLAH (Alwajih et al., 2024)	2024	6 Arabic dialects	MENA	✗	✓	✓	ArabicNLP
ARABIC DIALECT DIALOGUE (Naous et al., 2020)	2020	Levantine, Egyptian, Gulf	MENA	✓	✗	✓	WANLP
ARABICAQA (Abdallah et al., 2024)	2024	Arabic (MSA)	MENA	✗	✗	✗	SIGIR
MENA SPEECH BANK (Ali et al., 2026)	2026	Arabic + dialects	MENA	✓	✓	✓	arXiv
SHAWARMA CHATS (Zeinalipour et al., 2025)	2025	MSA, Egyptian, Maghrebi	MENA	✓	✗	✓	ArabicNLP
ALEXANDRIA (El Mekki et al., 2026)	2026	en + 13 Arabic dialects	13 Arab countries	✓	✗	✓	ACL
<i>Chinese</i>							
ALIGNMMBENCH (Wu et al., 2025d)	2025	zh	China	✓	✓	✗	ACL
CMT-EVAL (Tian et al., 2025a)	2025	zh	China	✓	✗	✓	arXiv
FB-BENCH (Li et al., 2025d)	2025	zh + en	China	✓	✗	✗	EMNLP
<i>Multilingual</i>							
CULTURALGROUND (Nyandwi et al., 2025)	2025	39 langs	42 countries	✗	✓	–	EMNLP
CVQA (Mogrovejo et al., 2024)	2024	26 langs	30 countries	✗	✓	–	NeurIPS
MMA-ASIA (Weihua et al., 2025)	2025	10 langs (8 Asian countries)	Asia	✗	✓	–	arXiv
MSTEB (Beyene et al., 2025)	2025	40+ langs	–	✗	✓	–	arXiv
M2LINGUAL (Maheshwary et al., 2025)	2025	ML	–	✓	✗	–	arXiv
WILDCHAT (Zhao et al., 2024b)	2024	ML (incl. code-switch)	200+ countries	✓	✗	–	ICLR
MLQA (Lewis et al., 2020)	2020	en, ar, de, es, hi, vi, zh	Cross-lingual	✗	✗	–	ACL
ADU-BENCH (Gao et al., 2025a)	2025	ar, zh, en, fr, de, ja, ko, ru, es	Multilingual audio dialogue	✗	✓	✗	ACL
MT-BENCH-HI (Kamath et al., 2025)	2025	Hindi	India	✓	✗	✗	ACL-W
CU-DIALOG (Cao et al., 2024)	2024	en	Multiple cultures	✓	✗	✗	EACL-F
INDOTOD (Kautsar et al., 2023)	2023	Indonesian	Southeast Asia	✓	✗	✗	SEALP
TYPHOON-AUDIO (Manakul et al., 2025)	2025	Thai, English	Thailand / Thai speech	✗	✓	✗	Interspeech
SEA-VQA (Uraierprasert et al., 2024)	2024	en	Southeast Asia / 8 countries, 53 cultures	✗	✓	✗	ALVR

Table 4: **Cultural and linguistic resources relevant to multi-turn / multimodal dialogue.** MT: ✓ = multi-turn structure; ✗ = single-turn only. MM: ✓ = more than one modality (vision and/or speech). Dialect: ✓ = dialect-aware (beyond MSA or a single variety).

dialogue benchmark with native-speaker translations of CamRest and SMD, covering NLU, dialogue state tracking, and response generation.

Single-turn cultural and linguistic anchors.

OASIS, developed through the EverydayMMQA framework (Alam et al., 2025), is a culturally grounded spoken visual QA resource covering English, MSA, and Arabic dialectal varieties across 18 Arab countries. It combines real images, text questions, spoken questions, and image-grounded answers, but its current task format is single-turn rather than conversational. M2CQA (Mousi et al., 2026) provides a single-turn multimodal QA benchmark spanning 17 MENA countries in MSA and multiple Arabic dialects, and introduces the Cultural Hallucination and Factual Recall metric for measuring culturally grounded correctness. ArabicaQA (Abdallah et al., 2024) supports Arabic question answering, Dallah (Alwajih et al., 2024) focuses on dialect-aware Arabic multimodal modeling, MLQA (Lewis et al., 2020) provides a cross-lingual extractive QA baseline, and mSTEB (Beyene et al., 2025) extends multilingual evaluation to speech and text tasks.

Broader cultural multimodal resources. SEA-VQA (Uraierprasert et al., 2024) evaluates cultural visual question answering across Southeast Asian settings. CVQA (Mogrovejo et al., 2024), CulturalGround (Nyandwi et al., 2025),

and MMA-ASIA (Weihua et al., 2025) extend culturally focused multimodal evaluation across broader language and regional settings. These resources are mostly single-turn, but they provide useful baselines and design signals for future culturally grounded multi-turn dialogue benchmarks.

A.2 Modeling Paradigms and Training Strategies

A.2.1 Classical Dialogue Systems

Classical dialogue systems treated dialogue as a pipeline of separate modules: natural language understanding, dialogue state tracking, policy learning, and natural language generation. Early rule-based systems used hand-written templates and finite-state managers. Statistical approaches later modeled dialogue as a partially observable Markov decision process (Gašić et al., 2013). Neural sequence-to-sequence models (Sutskever et al., 2014; Serban et al., 2016) replaced parts of the response-generation pipeline with end-to-end models, but the broader system design often remained modular. These systems remain useful baselines for understanding how later Transformer- and LLM-based dialogue models absorbed functions that were previously handled by separate components.

A.2.2 Transformer Dialogue Architectures

Pre-LLM Transformer dialogue architectures established many capabilities that were later sub-

Method	Year	Type	MT-aware	Multi-turn novelty	Venue
(i) Supervised fine-tuning					
WILDCHAT (Zhao et al., 2024b)	2024	SFT	✓	1M real opt-in convs; multilingual	ICLR
PARROT (Sun et al., 2024)	2024	SFT+PO	✓	Parrot-Ask elicits anaphora/ellipsis in follow-ups; SFT + context-aware preference opt.	ACL
AQUILA-MED (Zhao et al., 2024a)	2024	SFT+RLHF+DPO	✓	Full-process medical dialogue recipe	arXiv
QILIN-MED (Ye et al., 2024a)	2024	SFT+DPO	✓	Multi-stage knowledge injection for medical multi-turn dialogue	arXiv
ZHONGJING (Yang et al., 2024)	2024	SFT+RLHF	✓	Expert-feedback RLHF + real-world multi-turn medical dialogue	AAAI
(ii) Reinforcement learning & preference optimisation					
INSTRUCTGPT (Ouyang et al., 2022)	2022	PPO	✗	RLHF preference recipe	NeurIPS
DMPO (Shi et al., 2024)	2024	DPO (MT)	✓	Direct multi-turn preference optimisation	EMNLP
M-DPO / M-KTO (Xiong et al., 2025)	2025	DPO/KTO	✓	Trajectory-level reward assignment (multi-turn math agents)	ICLR
ITPO (Wang et al., 2026)	2026	RL	✓	Implicit turn-wise policy optimisation	arXiv
SDPO (Kong et al., 2025)	2025	DPO (segment)	✓	Segment-level preference optimisation for social dialogue	ACL
DIATool-DPO (Jung et al., 2025)	2025	DPO	✓	Multi-turn dialogue-control optimisation for tool use	SIGDIAL
ARChER (Zhou et al., 2024b)	2024	HRL	✓	Two-level (on and off-policy RL)	ICML
REFUEL (Gao et al., 2025b)	2025	RLHF	✓	Efficient multi-turn RLHF; regresses relative-future rewards	ICML
SWEET-RL (Zhou et al., 2025)	2025	RL	✓	Turn-wise advantage critic for collaborative reasoning	arXiv
ACT (Chen et al., 2025e)	2025	DPO / self-training	✓	Action-level contrastive self-training (clarify vs. answer)	ICLR
SCoRE (Kumar et al., 2025)	2025	RL	✓	Online RL for self-correction over multi-turn traces	ICLR
MT-RLHF (Shani et al., 2024)	2024	RL	✓	Mirror-descent policy opt.; Nash from conversation-level prefs	NeurIPS
JOSH (Lattimer et al., 2025)	2025	Self-alignment/RL	✓	Sparse-reward self-training in MultiWOZ-derived tool env	ACL-F
DOCTORAGENT-RL (Feng et al., 2026)	2026	RL	✓	Multi-agent collaborative RL for multi-turn clinical dialogue	ICASSP
OFFLINE RL PERSONA (Shea and Yu, 2023)	2023	Offline RL	✓	Persona-break penalty, Persona consistency critic	EMNLP
STUDENT-SIM (Scarlato et al., 2025)	2025	SFT+DPO	✓	Tutor training via distillation + DPO; simulated-student model scores preference pairs	AIED
(iii) Multi-task learning					
DAMSEL (Chen et al., 2025d)	2025	Multi-task learning	✓	Auxiliary task design for spoken QA with limited speech data	ACL-F
(iv) Synthetic data generation					
ULTRACHAT (Ding et al., 2023)	2023	Synthetic data	✓	1.5M synthetic multi-turn convs via paired-LLM self-chat	EMNLP
MMDU-45K (Liu et al., 2024d)	2024	Synthetic (+bench)	✓	GPT-4o-generated multi-turn multi-image; doubles as benchmark	NeurIPS
TMDIALOG (Lei et al., 2025)	2025	Synthetic data	✓	GPT-4-generated multi-turn multimodal dialogue corpus	arXiv
MMDIAG (Liu et al., 2025a)	2025	Synthetic data	✓	GPT-assisted multi-turn multimodal dialogue with note annotations	EMNLP
ARABIC SFT CORPUS (Misbah et al., 2026)	2026	Synthetic data	✓	43K Arabic synthetic multi-turn dialogues; blueprint for Arabic SFT	PLO ONE
DEEPDIALOGUE (Koudounas et al., 2025)	2025	Synthetic data	✓	Synthetic multi-turn spoken dialogue with emotion-consistency control	arXiv
TOOLDIAL (Shim et al., 2025)	2025	Synthetic data	✓	Multi-turn dialogue generation pipeline for tool-augmented LMs	ICLR
CONSISTENTCHAT (Chen et al., 2025a)	2025	Synthetic data	✓	Skeleton-guided generation with nine intent trajectories	EMNLP
DocTALK (Lee et al., 2025a)	2025	Synthetic (PT)	✓	Graph-based synthesis of 730k multi-turn dialogues from Wikipedia (pre-training)	SIGDIAL
REVIEW-INSTRUCT (Wu et al., 2025c)	2025	Synthetic data	✓	Ask-Respond-Review three-role pipeline for high-consistency data	ACL-F
SELF-TALK (Ulmer et al., 2024)	2024	Synthetic data	✓	LLM-simulated user-agent task dialogue generation	ACL-F
APIGEN-MT (Prabhakar et al., 2026)	2026	Synthetic data	✓	Verifiable multi-turn agent trajectory generation	NeurIPS
(v) Conversational retrieval-augmented training					
CHATQA (Liu et al., 2024c)	2024	SFT (retrieval)	✓	Two-stage tuning of retriever + LLM for multi-turn conversational QA	NeurIPS
CHATQA-2 (Xu et al., 2025c)	2025	SFT (retrieval)	✓	Three-stage context-expansion tuning (8K128K) for long conversational RAG	ICLR
HACONVDR (Mo et al., 2024)	2024	Dense retrieval	✓	Context-dennoised reformulation + turn-impact supervision	ACL-F
CONVAUG (Chen et al., 2024a)	2024	Dense retrieval	✓	LLM-cognition augmentation + difficulty-adaptive contrastive learning	ACL

Table 6: **Multi-turn training strategies**, grouped by the five families of §5. **Type:** SFT=supervised fine-tuning; DPO/KTO=direct-preference / Kahneman–Tversky optimisation (MT=multi-turn, segment=segment-level); RLHF/PPO, RL, HRL=hierarchical RL; *retrieval*=conversational-retrieval training; *Synthetic*=data-generation pipeline; Combined types (e.g. SFT+DPO) mark hybrids that span families; each work is listed under its *primary* contribution. **MT-aware:** ✓if the method explicitly optimises across turns rather than treating each turn independently.

filling, dialogue state tracking, policy prediction, and response generation. A related line incorporated structured knowledge and dialogue state tracking. Lin et al. (2021b) inject schema and knowledge-graph structure into GPT-2 for dialogue state tracking, while Feng et al. (2023b) show that LLMs can directly track slot-value pairs across turns without a dedicated DST module. Another line addresses long context and memory. Transformer-XL (Dai et al., 2019) reuses hidden states across segments, Recurrent Memory Transformer (Bulatov et al., 2022) adds recurrent memory tokens, RWKV (Peng et al., 2023) reformulates attention as a linear recurrence, and MemBART (Wu and Yu, 2024) adds a gated memory module to preserve dialogue history without simply enlarging the input window.

A.2.3 LLM-based Dialogue Models

The LLM-based phase shifted dialogue modeling away from task-specific managers toward general-

purpose instruction-following models. Instruct-GPT (Ouyang et al., 2022) helped define this phase through supervised fine-tuning, reward modeling from human preferences, and reinforcement learning from human feedback. FnCTOD (Li et al., 2024c) shows that task-oriented structure can be recovered inside an LLM by treating each dialogue slot as a callable function and letting the model fill slot values zero-shot. LLM-centric systems for conversational QA and RAG, including ChatQA, ChatQA-2, and UniConv, are discussed in Section 5, since their primary contribution is training rather than architecture.

A.2.4 AudioLLMs

AudioLLMs extend conversational models to speech by jointly modeling language and audio. Early systems such as SpeechGPT and AudioPaLM represent speech through discrete units or audio tokens within an LLM, while Qwen-Audio, Qwen2-Audio, SALMONN, Kimi-Audio,

Strategy	Training signal	Role across turns	Best suited to	Key trade-off	Modality
<i>Objective-level strategies</i>					
Supervised fine-tuning	Reference dialogue turns; token-level likelihood	Implicit: learns responses conditioned on dialogue history	General ability; dialogue domain, format, and safety adaptation	Stable and efficient, but provides no explicit delayed or cross-turn credit	T / S / MM
Multi-task learning	Dialogue loss with auxiliary objectives such as persona, knowledge, retrieval, or dialogue state	Indirect: auxiliary tasks encourage consistency and grounding	Persona and knowledge grounding; dialogue-state tracking; transfer across skills	Improves transfer, but may cause negative transfer and requires loss balancing	T / MM
Preference mization	Preference labels or rewards over responses or trajectories	Explicit: can optimize outcomes over multiple turns	Preference alignment; long-horizon task success; safety	Supports delayed credit, but can be costly and sensitive to reward misspecification	T
<i>Data and grounding strategies</i>					
Synthetic data	Generated, self-play, or distilled multi-turn dialogues	Coverage: expands the diversity and length of dialogue trajectories	Low-resource settings; rare behaviors; safety and robustness cases	Scalable and controllable, but may amplify errors and biases and requires filtering	T / (S, MM)
Conversational RAG	History-aware retrieval and grounded-response supervision	Grounding: teaches evidence use across dialogue turns	Knowledge-intensive dialogue; long histories; changing knowledge	Improves grounding, but depends on retrieval quality and adds latency	T / (MM)

Table 7: Comparison of **strategies for training multi-turn conversational systems**. The role across turns indicates how each strategy supports learning beyond isolated turns. T = text; S = speech; MM = multimodal; parentheses indicate emerging use. Strategies are complementary rather than mutually exclusive.

and SpiRit-LM broaden this direction toward audio understanding, spoken dialogue, and cross-modal learning (Zhang et al., 2023; Rubenstein et al., 2023; Chu et al., 2023, 2024; Tang et al., 2024; Team, 2024; Nguyen et al., 2025).

A second line focuses on real-time and full-duplex spoken interaction. Moshi supports low-latency full-duplex dialogue and uses time-aligned text tokens as an inner monologue before speech generation (Défossez et al., 2024). Subsequent systems explore streaming speech generation, controllable voice, style and timbre modeling, and simultaneous listening and speaking (Chen et al., 2025g; Fang et al., 2025; Li et al., 2024b; Zeng et al., 2024; Xiezhifei, 2024; Chen et al., 2025h). Freeze-Omni keeps the LLM backbone frozen while training speech input and output modules, whereas SALMONN-omni jointly models user speech and system output in a codec-free architecture that supports turn-taking, barge-ins, and echo cancellation (Wang et al., 2025b; Yu et al., 2026).

Other work separates duplex control from the dialogue backbone. FlexDuo uses a Speak-Listen-

Idle state machine to reduce false interruptions, while FireRedChat combines personalized VAD with semantic end-of-turn detection for controllable barge-in (Liao et al., 2025; Chen et al., 2025b). In a different direction, dGSLM models two-channel raw audio without text supervision, generating speech, laughter, and other paralinguistic signals (Nguyen et al., 2023). Lu et al. (2026) provide a broader review of full-duplex architectures, interaction ontologies, and decision states.

Overall, AudioLLMs have substantially improved speech-native and real-time interaction. However, most systems still rely on the underlying language model to maintain dialogue history, with limited explicit modeling of long-horizon spoken memory and cross-turn acoustic grounding.

A.2.5 Omni-modal Models

Omni-modal models extend AudioLLMs by integrating vision with text and speech, and in some cases video. GPT-4o provides an end-to-end omni-modal system with real-time streaming, while Mini-Omni2 extends Mini-Omni with

visual input (Hurst et al., 2024; Xie and Wu, 2024; Xiezhifei, 2024). Other systems explore low-latency vision-speech interaction, simultaneous multimodal processing, unified tokenization, and multimodal pretraining (Fu et al., 2026; Chen et al., 2025c; Zhang et al., 2025c; Guo et al., 2025; Wang et al., 2025d). Royer et al. (2025) directly connect speech and vision encoders without using text as an intermediate representation.

Recent models further broaden this design space through instruction tuning, multimodal generation, video-audio alignment, and long-horizon speech modeling (Li et al., 2024a; AI et al., 2025; Xu et al., 2025b; Wang et al., 2025a). Audio-video models such as VideoLLaMA 2 and CAT additionally target reasoning over dynamic audio-visual content (Cheng et al., 2024; Ye et al., 2024b).

A separate line focuses explicitly on multi-turn interaction. Most omni-modal models treat dialogue history as a flat input sequence, whereas newer methods introduce mechanisms for memory and context management. DiagNote uses explicit note-taking, CoLVLM follows a memory-perception-planning-execution loop, and ContextQFormer maintains a dedicated contextual memory block (Liu et al., 2025a; Han et al., 2025; Lei et al., 2025). LoopServe and MadaKV reduce inference cost in long multimodal sessions, while DialogGen conditions image generation and editing on conversational history (Li et al., 2025a,c; Huang et al., 2025b).

Collectively, omni-modal models have broadened multimodal interaction; however, explicit cross-turn memory and long-horizon grounding remain less developed.

A.2.6 Agentic and Tool-augmented Dialogue

Agentic dialogue systems use conversation as an interface for planning, tool use, observation, and revision. ReAct interleaves reasoning with external actions, Reflexion stores verbal feedback from failed attempts for later use, and MetaGPT coordinates multiple role-based LLM agents through structured messages (Yao et al., 2022; Shinn et al., 2023; Hong et al., 2024).

Tool-augmented systems connect this process to external environments. ToolPlanner combines path planning with process- and outcome-level feedback, while ChatCoT integrates external tools into reasoning (Wu et al., 2024; Chen et al., 2023). τ -Bench evaluates agents in retail and airline settings using APIs and simulated users, whereas

ToolSandbox focuses on stateful tool use over variable-length interactions (Yao et al., 2025; Lu et al., 2025). xLAM-2 represents open models for multi-turn function calling (Prabhakar et al., 2026).

Other systems extend agentic interaction to web navigation, recommendation, video, and streaming environments. WebLINX evaluates conversational web navigation, while SAPIENT applies Monte Carlo Tree Search to conversational recommendation (Lu et al., 2024; Du et al., 2025a). Zhang et al. (2024d) incorporate users’ emotional states into recommendation. VideoMind and Video-MTR support iterative reasoning over video, while IXC2.5-OmniLive combines streaming perception, multimodal long-term memory, and reasoning for live interaction (Liu et al., 2026; Xie et al., 2025; Zhang et al., 2024c).

Overall, agentic systems highlight three multi-turn requirements: tracking external state across actions, replanning as goals or environments change, and maintaining memory over extended interactions.

A.2.7 Supervised Fine-tuning

Supervised fine-tuning adapts models to multi-turn interaction using conversations that preserve cross-turn dependencies. UltraChat provides large-scale synthetic instructional dialogues, while WildChat contains real opt-in conversations with broad topic and language coverage. Parrot explicitly targets anaphora and ellipsis in follow-up turns (Ding et al., 2023; Zhao et al., 2024b; Sun et al., 2024). Domain-specific work extends SFT to clinical dialogue and adversarial safety (Zhao et al., 2024a; Ye et al., 2024a; Yang et al., 2024; Rahman et al., 2025). SFT provides a practical foundation for multi-turn training, with performance largely shaped by the coverage and quality of training data.

A.2.8 Reinforcement Learning and Preference Optimization

Multi-turn reinforcement learning and preference optimization address delayed rewards and credit assignment across turns. InstructGPT established the standard RLHF pipeline, while ArCher and MT-RLHF extend optimization toward turn- and conversation-level feedback (Ouyang et al., 2022; Zhou et al., 2024b; Shani et al., 2024). Preference-based methods such as DMPO, Multi-turn DPO/KTO, Parrot, and SDPO further opti-

mize dialogue histories, interaction sequences, or segments rather than individual responses (Shi et al., 2024; Xiong et al., 2025; Sun et al., 2024; Kong et al., 2025).

A second line focuses on actions and outcomes across turns. Action-Based Contrastive Self-Training distinguishes actions such as answering and clarifying, while DiaTool-DPO and JOSH optimize tool-use behavior from interaction sequences (Chen et al., 2025e; Jung et al., 2025; Lattimer et al., 2025). Other approaches target self-correction, collaborative reasoning, proactive interaction, and sparse long-horizon rewards (Kumar et al., 2025; Zhou et al., 2025; Gao et al., 2025b; Wang et al., 2026; Abdulhai et al., 2025). Domain-specific work further applies these methods to clinical and tutoring dialogue (Scarlatos et al., 2025; Feng et al., 2026).

More broadly, these methods extend optimization from individual responses to cross-turn behavior and long-horizon outcomes.

A.2.9 Multi-task Learning

Multi-task learning improves dialogue modeling by sharing representations across related objectives. PPTOD jointly trains response generation, dialogue state tracking, and policy prediction, while TOD-BERT learns dialogue-aware representations across multiple task-oriented dialogue corpora (Su et al., 2022; Wu et al., 2020). DAMSEL extends this approach to spoken conversational QA, using auxiliary objectives to improve learning when labeled speech data is limited (Chen et al., 2025d).

Taken together, multi-task learning improves transfer across related dialogue skills, although most approaches do not explicitly optimize session-level dependencies.

A.2.10 Synthetic Data Generation

Synthetic data generation scales multi-turn training data while allowing control over dialogue structure, topic, and modality. UltraChat simulates user-assistant conversations, while MMDU-45K, MMDiag, and TMDialog extend synthetic generation to multimodal dialogue, note-taking, and context modeling (Ding et al., 2023; Liu et al., 2024d, 2025a; Lei et al., 2025). Other pipelines target Arabic and emotional spoken dialogue, tool use, coherent intent sequences, and multi-topic information seeking (Misbah et al., 2026; Koudounas et al., 2025; Shim et al., 2025; Chen et al., 2025a;

Lee et al., 2025a). Recent approaches also use reviewer-guided regeneration, task-oriented dialogue bootstrapping, and simulated API interactions to improve data quality and task coverage (Wu et al., 2025c; Ulmer et al., 2024; Prabhakar et al., 2026).

In summary, synthetic generation offers a scalable way to target multi-turn behaviors, shifting from general dialogue scaling toward targeted simulation across modalities and tasks.

A.2.11 Training for Conversational RAG

Conversational RAG trains models to retrieve and use evidence as dialogue context evolves. PK-ICR jointly retrieves persona and external knowledge, while commonsense retrieval helps fill knowledge gaps in open-domain dialogue (Oh et al., 2023; Varshney et al., 2022). ChatQA and ChatQA-2 jointly train retrieval and generation for conversational and long-context RAG (Liu et al., 2024c; Xu et al., 2025c). Other methods improve history selection, retrieval supervision, and query reformulation: HAConvDR filters noisy dialogue history, UniConv jointly optimizes retrieval and generation, ConvAUG strengthens retrieval through LLM-cognition data augmentation, and IterCQR iteratively reformulates queries using retrieval feedback (Mo et al., 2024, 2025; Chen et al., 2024a; Jang et al., 2024). KEDiT compresses retrieved evidence into adapter parameters, while CORAL incorporates citation labeling into conversational RAG training and evaluation (Zhang et al., 2025a; Cheng et al., 2025).

Overall, conversational RAG has progressed toward history-aware retrieval and evidence integration across turns. In Table 8, we map common multi-turn training goals to the primary strategies discussed above, providing a practical guide for strategy selection.

A.3 Details on Evaluation

In Table 9, we summarize representative metrics and frameworks across the five evaluation families. In Table 10, we compare these families by their evaluation focus, suitable settings, and main limitations. We discuss each family in detail below.

A.3.1 Evaluation Families

Surface-form and task-oriented metrics. Surface-form metrics such as BLEU, ROUGE, and BERTScore remain useful baselines; however, they mainly score individual responses and

Training goal	Primary strategy
General conversational and instruction-following ability	SFT on curated multi-turn dialogues
Domain-specific or safety-focused behavior	SFT with targeted synthetic augmentation
Human preference and long-horizon outcomes	Preference optimization or multi-turn RL
Knowledge-intensive multi-turn QA	Conversational RAG training
Low-resource settings or rare behaviors	Synthetic data generation
Persona/knowledge grounding and skill transfer	Multi-task learning

Table 8: Mapping common multi-turn training goals to their primary training strategies. These strategies can also be combined depending on the target setting.

provide limited insight into cross-turn behavior. They do not capture whether a model preserves instructions, persona, grounding, or coherence across a session. Task-oriented metrics such as joint goal accuracy, Slot-F1, Inform, and Success better reflect multi-turn task completion by tracking dialogue states and outcomes. However, they remain task-specific, can be sensitive to paraphrasing, and often reduce success to binary outcomes. MM-Relevance extends response relevance to image-text dialogue, although it is limited to two-modal settings.

Session-level dialogue metrics. Session-level metrics directly assess behaviors that depend on earlier turns, including memory recall, instruction retention, constraint satisfaction, feedback integration, and dialogue-level hallucination. Representative approaches measure rubric compliance (APR and ARS), positional consistency (PWC), multimodal memory and reasoning (MMRC), interaction patterns (MT-Eval), structural constraints (WCSR), instruction hierarchy (IHEval), programmatic instruction following (PIF), feedback-based reasoning (TurnBench-MS), and dialogue-level hallucination (DiaHalu). Together, these metrics provide a broader view of cross-turn competence, although their definitions and scoring remain fragmented across benchmarks.

Agentic and retrieval-grounded evaluation.

Agentic and retrieval-grounded systems require evaluation of both responses and actions across an interaction. A fluent answer can still fail if the model uses an incorrect tool state, calls an API with missing information, or cites the wrong source. CORAL jointly measures retrieval, gen-

eration, and citation attribution, while τ -Bench measures repeated-trial task success through τ -pass^k. ToolSandbox evaluates stateful interactions using milestone and minefield scoring, and AgentBoard tracks progress through subgoal completion. These settings make external state, side effects, and repeated-trial reliability central evaluation concerns.

Speech-native and full-duplex evaluation.

Spoken dialogue evaluation must consider both semantic content and interaction dynamics. ADU-Bench covers open-ended audio dialogue across skills, languages, and ambiguity types, while MTalk-Bench combines pairwise and rubric-based scoring for semantic content, vocal cues, and ambient sound. ASK-QA focuses on spoken clarification, and URO-Bench separates correctness, speech quality, speech-text matching, and first-packet latency. FD-Bench adds interruption and timing measures for full-duplex interaction, while Full-Duplex-Bench-v3 extends evaluation to tool use and disfluency handling. These benchmarks show that transcript correctness alone is insufficient for evaluating spoken interaction.

LLM-as-a-judge and human evaluation.

LLM-as-a-judge and human evaluation support dimensions that are difficult to score automatically, including open-ended coherence, persona consistency, multimodal grounding, and human-likeness. MT-Bench, GPTScore, JudgeLM, BotChat, ContextualJudgeBench, LLM-Eval Analysis, and Multi-Judge Evaluator represent judge-based approaches, while ABC-Eval, MMDU, MTalk-Bench, and WildBench incorporate human or hybrid assessment. These methods offer flexible evaluation, although their reliability depends on rubric design and can be affected by judge bias, rater inconsistency, cost, and small ranking differences.

Overall, these evaluation methods capture complementary aspects of multi-turn performance, but no single approach fully measures session-level competence across modalities and interaction settings.

A.4 Search Keywords

We organized the literature search into six query groups covering core multi-turn dialogue, spoken and audio interaction, multimodal and omnimodal dialogue, training and alignment, evaluation, and cultural and multilingual settings. Within

Metric / Framework	Type	What it measures	MT-spec.	Core limitation	Applied in
Surface-form and task-oriented					
BLEU (Papineni et al., 2002)	NLG	n-gram precision	✗	No semantics; no coherence signal	MT-Bench, CoQA
ROUGE (Lin, 2004)	NLG	n-gram recall	✗	Surface-form only; ignores context	SpokenWOZ
BERTScore (Zhang et al., 2020a)	NLG	Semantic embedding similarity	✗	Single utterance; history-blind	MMDU
MM-Relevance (Feng et al., 2023a)	NLG	Modality-aware response relevance	✓	Defined for two-modal only	MMDialog
JGA	DST	Joint goal accuracy across slots	✓	Brittle to surface paraphrase	MultiWOZ
Slot-F1	DST	Per-slot precision and recall	✗	Slot-type bias; misses long-range context	SpokenWOZ
Inform / Success	DST	Entity provision and task success	✓	Binary; misses partial success	MultiWOZ
Session-level metrics					
APR & ARS (Deshpande et al., 2025; Gosai et al., 2025)	CONS	Average pass rate and rubric score	✓	Turn-level and session-level scores can diverge	MultiChallenge
PWC (Li et al., 2025e)	CONS	Position-weighted consistency	✓	Requires adversarial probe set	MT-Eval style
MMRC 6-axis (Xue et al., 2025)	MEM	Extract, reason, update, manage, recall, refuse	✓	Image management is rare in text-only settings	MMRC
TURNWISE gap (Graf et al., 2026)	CONS	Matched single-turn and multi-turn gap	✓	Requires paired single-turn references	General LLMs
WCSR (Li et al., 2025b)	CONS	Structural and intra-turn constraint satisfaction	✓	Depends on constraint extraction and LLM judging	StructFlowBench
MT-Eval patterns (Kwan et al., 2024)	CONS	Recollection, expansion, refinement, follow-up	✓	Single-session; no cross-session coverage	MT-Eval
IHEval (Zhang et al., 2025e)	CONS	Instruction hierarchy across system, user, history, and tool outputs	✓	Conflict resolution remains difficult	IHEval
Feedback-loop rule inference (Zhang et al., 2025d)	CONS	Integration of structured feedback across turns	✓	Game setting may not cover open-domain dialogue	TurnBench-MS
DiaHalu taxonomy (Chen et al., 2024b)	CONS	Dialogue-level hallucination subtypes	✓	Subtype labels require human annotation	DiaHalu
PIF (Epstein et al., 2024)	CONS	Programmatic instruction following across accumulated constraints	✓	Checks format constraints, not answer correctness	MMMT-IF
Agentic and retrieval-grounded					
CORAL citation labeling (Cheng et al., 2025)	AGENT	Retrieval, generation, and citation attribution	✓	Citation annotation requires human labeling	CORAL
r-pass ⁶ (Yao et al., 2025)	AGENT	Task pass rate across repeated trials	✓	Requires live user simulator; high cost	r-Bench
ToolSandBox scoring (Lu et al., 2025)	AGENT	Stateful trajectory with milestone and minefield scoring	✓	Simulator-dependent; limited domains	ToolSandBox
AgentBoard progress rate (Ma et al., 2024)	AGENT	Subgoal completion across agent trajectories	✓	Requires task-specific progress decomposition	AgentBoard
Speech-native and full-duplex					
ADU-Bench (Gao et al., 2025a)	JUDGE	Open-ended audio dialogue skills and ambiguities	✓	Judge-dependent; rubric design central	ADU-Bench
MTalk-Bench protocol (Du et al., 2025b)	JUDGE+HUMAN	Pairwise and rubric-based S2S scoring	✓	Small ranking gaps unstable	MTalk-Bench
ASK-QA outcome similarity (Chen et al., 2025d)	CONS	Semantic similarity after spoken clarification turns	✓	Uses simulated users and TTS follow-ups	ASK-QA
URO-Bench (Yan et al., 2025b)	CONS	Correctness, speech quality, speech-text match, latency	✓	Dimensions are not always separable	URO-Bench
FD-Bench interruption set (Peng et al., 2025)	FD	SIR, SRIR, EIR, NIR, and SRR	✓	Synthetic user speech; limited systems	FD-Bench
FD-Bench timing set (Peng et al., 2025)	FD	IRD, FSED, ERT, and EIT	✓	Requires reliable VAD, ASR, and audio logs	FD-Bench
Full-Duplex-Bench-v3 (Lin et al., 2026)	FD+AGENT	Tool-use correctness and disfluency tolerance	✓	Emerging full-duplex tool-use setting	Full-Duplex-Bench-v3
LLM-as-judge and human evaluation					
MT-Bench (Zheng et al., 2023)	JUDGE	Pairwise preference and 1-10 rating	✓	Self-enhancement and verbosity bias	MT-Bench
GFTScore (Fu et al., 2024)	JUDGE	Scoring through instruction-prompt likelihood	✓	Prompt-sensitive; weak dialogue grounding	Multi-turn dialogue quality
LLM-Eval Analysis (Zhang et al., 2024a)	JUDGE	Coherence, engagement, informativeness	✓	No multi-turn-specific axes	Open-domain dialogue
BotChat (Duan et al., 2024)	JUDGE	Human-likeness discrimination	✓	Confounded by GPT-4 dominance	Open LLMs
JudgeLM (Zhu et al., 2025)	JUDGE	Fine-tuned open judge	✓	Knowledge and position bias	JudgeLM bench
ContextualJudgeBench (Xu et al., 2025a)	JUDGE	Judge consistency under context and RAG	✓	Contextual sensitivity; reference dependence	RAG + multi-turn eval
Multi-Judge Evaluator (Tang et al., 2025)	JUDGE	Distilled agreement from multiple judges	✓	Dependent on judge pool quality	Multi-turn dialogue
ABC-Eval (Zhao et al., 2026a)	HUMAN	Fine-grained behavioural criteria	✓	Labour-intensive; rater drift	Open-domain chat
MMDU rubric judging (Liu et al., 2024d)	HUMAN	Long-form multimodal rubric judgment	✓	Expensive; hard to scale	MMDU
WildBench (Lin et al., 2025)	JUDGE	Pairwise evaluation on real user conversations	✓	Scores individual responses pairwise	WildBench

Table 9: **Representative evaluation metrics and frameworks** for multi-turn dialogue. **Type:** NLG = surface-form overlap; DST = task-state tracking; CONS = consistency; MEM = memory; AGENT = agentic/stateful; JUDGE = LLM-as-judge; HUMAN = human rubric; FD = full-duplex speech. **MT-spec.:** ✓ if the metric is undefined or degenerate in single-turn settings.

FD-Bench metric abbreviations: SRR = successful reply rate; SIR = successful interrupt rate; SRIR = successful reply-to-interrupt rate; EIR = early interrupt rate; NIR = noise interrupt rate; IRD = interrupt response delay; FSED = first speech emit delay; ERT = early reply time; EIT = early interrupt time.

each group, OR connected alternative keywords, while AND combined complementary concepts. This strategy aimed to provide broad coverage across modalities, methods, benchmarks, and language settings.

Core multi-turn dialogue. We used “multi-turn”, “multi turn”, “multiturn”, “multi-round”, “conversational AI”, “dialogue system”, “dialog system”, “task-oriented dialogue”, “task oriented dialog”, “open-domain dialogue”, “chit-chat”, “conversational agent”, “dialogue state tracking”, “conversational history”, “context-aware dialogue”, and “dialogue management”. These terms were combined with “language model”, “LLM”, “large language model”, “transformer”, “BERT”, “GPT”, “T5”, “instruction tuning”, “fine-tuning”, “pre-trained model”, and “neural dialogue”.

Spoken and audio dialogue. Speech-related keywords included “spoken dialogue”, “speech dialogue”, “voice assistant”, “spoken language understanding”, “SLU”, “spoken QA”, “end-to-end spoken dialogue”, “speech-to-speech”, “spoken conversational”, “audio dialogue”, “ASR dialogue”, “automatic speech recognition”, “text-to-speech”, “TTS”, “speech synthesis”, “spoken

language model”, and “audio language model”. We combined these with “multi-turn”, “conversational”, “dialogue”, or “multi-round”.

Multimodal and omni-modal dialogue. We used “multimodal dialogue”, “visual dialogue”, “visual QA dialogue”, “video dialogue”, “image-grounded conversation”, “vision-language dialogue”, “VQA multi-turn”, “omni-modal”, “any-to-any”, “speech-vision”, “multimodal conversational agent”, “embodied dialogue”, “audio-visual dialogue”, “multi-modal chat”, and “visual chatbot”. We also included representative model families such as “LLaVA”, “InstructBLIP”, “Flamingo”, “GPT-4V”, “Gemini”, “Qwen-VL”, and “CogVLM”. These terms were combined with “multi-turn”, “conversational”, “context”, or “history”.

Training and alignment. Training-related keywords included “instruction tuning”, “instruction following”, “reinforcement learning from human feedback”, “RLHF”, “direct preference optimisation”, “DPO”, “LoRA”, “low-rank adaptation”, “QLoRA”, “parameter-efficient fine-tuning”, “PEFT”, “alignment”, “constitutional AI”, “RLAIF”, “context window”, “long context”,

Evaluation family	What it measures	Best suited for	Main limitation
Surface-form & task-oriented	Response overlap, semantic similarity, dialogue-state accuracy, and task success	Task-oriented dialogue and fast automatic scoring	Limited signal on session-level behavior and sensitivity to valid paraphrases
Session-level	Instruction retention, memory, constraint satisfaction, feedback integration, and dialogue-level hallucination	Memory, consistency, and long-horizon interaction	Metric definitions vary across benchmarks, limiting direct comparison
Agentic & retrieval-grounded	Tool use, stateful execution, evidence retrieval, and source attribution	Tool-augmented agents and conversational RAG	Limited support for hidden state, side effects, and repeated-trial reliability
Speech-native & full-duplex	Semantic quality, speech quality, paralinguistic cues, interruption handling, latency, and timing	Spoken and full-duplex dialogue systems	Semantic correctness, speech quality, and interaction timing are often evaluated separately
LLM-as-judge & human	Rubric-based and pairwise judgments of open-ended response and dialogue quality	Open-ended or subjective evaluation	Sensitive to judge bias, rater inconsistency, cost, and reproducibility

Table 10: Comparison of five evaluation families for multi-turn dialogue.

Survey	Conceptual framing	Modality	MT	Cult.	Eval Δ	Key distinction
Yi et al. (2025)	LLM-based multi-turn dialogue; methods, tasks, resources, and evaluation	T	✓	✗	✓	Broad synthesis centered on text; no common framework for comparing session-level requirements across modalities.
Wang et al. (2023)	Language-model dialogue; evolution of task-oriented and open-domain systems	T	✓	✗	✗	Historical view of dialogue systems across language-model generations, primarily in text.
Zhang et al. (2025b)	Multi-turn LLM capabilities; instruction following, memory, planning, reasoning, and evaluation	T	✓	✗	✓	Capability-centered analysis with limited comparison across speech, vision/video, and omni-modal interaction.
Li et al. (2025f)	Multi-turn LLM interaction; task families and model-, memory-, and agent-based methods	T	✓	•	✓	Broad multi-turn taxonomy; multimodal session behavior is not a central comparison dimension.
Guan et al. (2026)	Conversational-agent evaluation; tools, memory, planning, and interaction quality	T + Tool	✓	✗	✓	Detailed agent-evaluation framework without joint analysis of data, models, training, and evaluation across modalities.
Zhang et al. (2024b)	Multimodal LLMs; architectures, training, model taxonomy, and benchmarks	T + V (+S)	✗	✗	✗	Broad multimodal coverage, with dialogue treated mainly as an application rather than the primary unit of analysis.
This survey	Complete multi-turn session; modality breadth $\times$ session-level competence across data, models, training, and evaluation	T + S + V + Vid + O + Tool	✓	✓	✓	Cross-modal analysis using common session-level capabilities, including memory, revision, grounding, state tracking, timing, safety, and cultural-linguistic competence.

Table 11: Comparison with closely related surveys by conceptual framing and scope. **Modality**: T = text, S = speech/audio, V = vision, Vid = video, O = omni-modal, Tool = tool-augmented. **MT**, **Cult.**, and **Eval Δ** indicate multi-turn, cultural-linguistic, and multi-turn evaluation coverage, respectively. • denotes partial coverage.

“retrieval augmented generation”, “RAG”, and “memory-augmented”. We combined these with “dialogue”, “conversational”, “multi-turn”, “chatbot”, or “assistant”.

Evaluation frameworks and metrics. Evaluation-related searches used “dialogue evaluation”, “conversation evaluation”, “turn-level evaluation”, “multi-turn evaluation”, “coherence”, “engagement”, “consistency”, “factual grounding”, “hallucination”, “faithfulness”, “dialogue benchmark”, “conversational benchmark”, “human evaluation dialogue”, “automatic evaluation dialogue”, “BLEU dialogue”, “BERTScore dialogue”, “perplexity dialogue”, “FED”, “USR”, “GRADE”, “DialEval”, and “CGPR”.

Cultural and multilingual grounding. We used “multilingual dialogue”, “cross-lingual dialogue”, “Arabic dialogue”, “Arabic conversational”, “low-resource dialogue”, “culturally grounded”, “cul-

tural dialogue”, “dialect”, “code-switching”, “multilingual QA”, “cross-cultural conversation”, “Arabic NLP”, “Arabic language model”, “Arabizi”, “cultural benchmark”, “cultural bias”, “cultural adaptation”, “Gulf dialect”, “MSA dialogue”, and “Arabic ASR”. These terms were combined with “multi-turn”, “dialogue”, “conversational”, “chatbot”, or “assistant”.

B Comparison with Related Surveys

We extend the comparison in Table 1 in Table 11, where we examine the conceptual framing, modality coverage, and key distinctions of closely related surveys. We show how our survey centers the complete multi-turn session and analyzes session-level competence across modalities, training, evaluation, and cultural-linguistic settings.